Advanced Techniques for EEG Processing - Laboratory Exercises

Exercise 8a – Time Domain Analysis: ERP waveforms and topographical maps (Within-Subject Analysis, i.e. Within Subject Averages (WSA))

Note: In this exercise, you are required to use the function `topoplot` available in one subdirectory of the toolbox EEGLAB. For this function to run, it must be included in the Matlab path. To this aim, immediately at the starting of the Matlab session, type `eeglab` in the command window (in this way, all subdirectories of EEGLAB are included in the path for the current Matlab session). Then, close the GUI and clear (`clear`) all workspace. As long as the Matlab session remains open, we can directly call `topoplot`.

The file `sub-035_PreprocessStep2.mat` and the file `sub-003_PreprocessStep2.mat` containes the EEG data of two subjects (subject id: 035 and 003) involved in a 3-stimulus oddball paradigm, after application of all preprocessing steps (steps of Exercise 1 and Exercise 7). Specifically, each file contains:

`X`: preprocessed EEG data organized as a 3-D matrix with size $n \times m \times q$ where $n(= 60)$ is the number of channels, $m(= 500)$ the number of time samples per epoch, and $q(= 195$ for subject 035 and $= 197$ for subject 003) the overall number of epochs for that subject. The epochs correspond to standard, target and distractor stimuli and are intermixed. Each epoch lasts 1 s: $-200\,ms$ pre-stimulus to $+800\,ms$ post-stimulus.

`ch_names`: a cell array containing the labels of the n electrodes.

`srate`: containing the value of the sampling rate (= $500\,Hz$).

`stim_types`: cell array (size $1 \times q$) indicating which applied stimulus ('standard', 'target', 'distractor') each epoch corresponds to.

Apply the following steps first to the data of one subject (loading the corresponding file .mat) and then to the data of the other subject (loading the corresponding file .mat). Observe the variability in the results from one subject to another and the common aspects, <u>especially looking at the topological maps</u>.

1) Load the file `sub-yyy_PreprocessStep2.mat` (`yyy=003 or 035`).

2) Apply the baseline correction to `X`: in each epoch and for each channel, compute the mean value over the entire pre-stimulus period (corresponding to the first 101 time samples in each epoch), and subtract it from all values of the channel inside the epoch. To this aim, the following instructions can be used:

```
baseidx=[1 101]; %indices of the baseline period from -200 ms to 0 ms (these are the first 200 ms = 101 samples in each epoch)

%efficient way to remove baseline
baseline=mean(X(:,baseidx(1):baseidx(2),:),2); % n x 1 x q
X_rmb=X-baseline; % X_rmb has size n x m x q %baseline removed trial by trial

X=X_rmb; %rewrite X; now X has the baseline removed
```

3) Separate the epochs based on the stimulus they correspond to, obtaining three 3-D matrices: `X_standard` (size $n \times m \times q_stand$), `X_target` (size $n \times m \times q_targ$), `X_distractor` (size $n \times m \times q_distr$), where q_stand, q_targ, q_distr are the number of epochs corresponding to the standard stimulus, target stimulus, and distractor stimulus. To this aim, the following instructions can be used (here referred to the standard condition):

```
idx_standard=find(strcmp(stim_types,'standard')); %find the indices of the epochs corresponding to the presentation of the standard stimulus

X_standard=X(:,:,idx_standard);  % X_standard is a 3D matrix with size n x m x q_stand where q_stand is the number of the standard epochs
```

4) Compute the Within-Subject Averages (WSA) at all electrodes, averaging across the epochs corresponding to the same stimulus condition, obtaining three 2-D matrices: `WSA_standard` (size $n \times m$), `WSA_target` (size $n \times m$), `WSA_distractor` (size $n \times m$).

5) Consider the following three channels: Fz (ch=12), Cz (ch=30), and Pz (ch=47). In the same figure, draw three subplots (one per row). In the first subplot (first row), plot the WSA waveform of Fz, as a function of time, for the standard, target and distractor. In the second subplot (second row), plot the WSA waveform of Cz, as a function of time, for the standard, target and distractor. In the third subplot (third row), plot the WSA waveform of Pz, as a function of time, for the standard, target and distractor. In each subplot use `ylim([-7 +7])`. Observe the differences across the three channels. The following instructions can be used (here referred to the first subplot):

```
time=([0:1:m-1]/srate)*1000; %in ms, from 0 to 1000 ms (200 ms is the time of stimulus presentation)
time=time-200; %in ms, so time starts from -200 ms, and t = 0 ms corresponds to stimulus presentation

ylimits=[-7 7];

ch_F=12; %Fz

figure
subplot(311)
plot(time,WSA_standard(ch_F,:),time,WSA_target(ch_F,:),time,WSA_distractor(ch_F,:),'linewidth',2);
ylim(ylimits)
set(gca,'xtick',[-200:100:800])
xlabel('time (ms)')
ylabel('\muV')
title(ch_names{ch_F})
grid
```

6) Topographical maps of the WSA at different times. Consider the time instants $0, 100, 200, 300, 400, 500, 600, 700\ ms$ post-stimulus (where 0 corresponds to stimulus onset). Generate a figure organized into 3×8 subplots, with one row per stimulus condition (standard, target, disctractor) and one column per time instant. Along the first row, plot the 8 topographical maps related to the standard condition by computing (and visualizing) the mean at each electrode over $100\ ms$ around each time instant (from $-50\ ms$ to $+50\ ms$). To plot each topographical map, use the eeglab function `topoplot.m` (see instructions at page 3). Along the second row, plot the topographical maps of the target WSA, and along the third row plot the topographical maps of the distractor WSA, using the same procedure as in the first row. Observe the differences across the three conditions. The following instructions can be used (here referred to the standard condition):

```
frames=[0:0.1:0.7]*srate+0.2*srate;   %frames contain the time samples
%corresponding to the time instants [0 0.1 0.2 0.3 0.4 0.5 0.6 0.7] s
%where 0 s corresponds to the stimulus presentation (0.2 s from the beginning of the epoch)
n_conditions=3; % number of conditions =3 (standard, target, distractor); this is the number of rows in the figure
n_frames=length(frames); % number of time instants; this is the number of columns in the figure
clim=[-7 +7]; %this are microvolts

titles = {'0 ms', '100 ms', '200 ms', '300 ms', ...
      '400 ms', '500 ms', '600 ms', '700 ms'};

figure
for i=1:n_frames
    subplot(n_conditions,n_frames,i)
    start=frames(i)-0.05*srate;
    stop=frames(i)+0.05*srate;
    ta_WSA_standard=mean(WSA_standard(:,start:stop),2);
```

```
    topoplot(ta_WSA_standard,'Standard-10-20-Cap60.locs','maplimits',clim);
    title(titles{i})
end
```

To draw the topographical maps relative to target condition in the second row, the same 'for' loop as before should be repeted, but in the suplot instruction `i` must be replaced by `i+8` (and by `i+16` in the third 'for' loop to draw the maps relative to the distractor condition in the third row)

Exercise 8b – Time Domain Analysis: ERP waveforms and topographical maps (Across-Subject Analysis, i.e. Grand Averages (GA))

The .mat file `WSA_allsubjects.mat` contains the Within Subject Averages (WSA) of all 25 ($= n_subj$) subjects who participated in the experiment. In particular, the file contains:

`WSA_standard_allsubj` (size $n \times m \times n_subj$). Inside the $i-th$ page, it contains the WSA of all channels of the i_th subject for the Standard condition ($n = 60; m = 500$).

`WSA_target_allsubj` (size $n \times m \times n_subj$). Inside the $i-th$ page, it contains the WSA of all channels of the i_th subject for the Target condition ($n = 60; m = 500$).

`WSA_distractor_allsubj` (size $n \times m \times n_subj$). Inside the $i-th$ page, it contains the WSA of all channels of the i_th subject for the Distractor condition ($n = 60; m = 500$).

`ch_names`: a cell array containing the labels of the n electrodes.

`srate`: containing the value of the sampling rate (= 500 Hz).

1) Load the file `WSA_allsubjects.mat`.

2) Compute the Grand Averages (GA) at all electrodes, averaging across subjects for the same stimulus condition, obtaining three 2-D matrices: `GA_standard` (size $n \times m$), `GA_target` (size $n \times m$), `GA_distractor` (size $n \times m$).

3) Replicate point 5) of Exercise 8a plotting the GA waverforms at the three channels Fz, Cz, Pz, using `ylim([-3 +3])`.

4) Replicate point 6) of Exercise 8a plotting the topographical maps of the GA at different times, using `clim = [-3 +3]`

(NOTE: use the same instructions provided in points 5 and 6 of Ex. 8a, replacing WSA_standard with GAr_standard; the same for the target and distractor condition)

<u>EEGLAB function `topoplot`</u>

In this exercise, call the function `topoplot` with the following input arguments

```
topoplot(Y,'file.locs','maplimits',clim)
```

`Y` is a $n \times 1$ vector containing the values electrode by electrode to be assigned to the electrode locations in the scalp map.

`'file.locs'` is the name of the file (between quotes) containing the locations of the electrodes. Here `'Standard-10-20-Cap60.locs'`

`'maplimits'` is a property of the map defining the range of the values ([min max] in µV) colorcoded in the map and `clim` is the value assigned to this property. For example, `clim = [-7 +7]`.